

SIT281 Cryptography - Task 2 (Week 9)

1. Discrete Logarithm, Diffie Hellman Key Exchange, ElGamal Scheme
 - a. Find x of $5^x = 9 \pmod{29}$ using Baby Step, Giant Step
 - b. Let $q = 29$ and $a = 9$. Take any two numbers as the private keys of two communicating parties, and show how a common key can be obtained by them as a shared secret key
 - c. Choose a decimal number as the message and show how the communication parties can encrypt/ decrypt the messages using the shared key in Part (b) with the same parameters
2. Secret Splitting and Sharing
 - a. Calculate all the points defined by the equation $y = (4x + 3) \pmod{7}$
 - b. Select any two points on the equation in Part (a) and show how a secret can be reconstructed using them
 - c. Calculate all the points defined by the equation $y = (x^2 + 5x + 3) \pmod{11}$
 - d. Select any three points on the equation in Part (c) and show how a secret can be reconstructed using them.
3. Elliptic Curve Cryptography
 - a. Show that the two points $(1,4)$ and $(4,2)$ lie on the elliptic curve $y^2 = x^3 + x + 1 \pmod{13}$
 - b. Add the two points calculated on the curve in Part (a) and show that it is on the graph defined in Part (a)